



EC 101B: Fundamentals of Object Oriented Programming

Course Code	EC 101B
Course Name	Fundamentals of Object Oriented Programming
L-T-P-C	3-0-2-4
Prerequisites	NIL Sem. Both
Approval	5th Senate (Oct 2013)

Course Contents

1. **Basic Computer Fundamentals:** Introduction to computer systems, computer as a programmed machine: CPU organization, ALU, registers, memory: machine language, assembly language, idea of program execution at micro level, high – level languages: concept of flow chart and algorithm algorithms to programs, efficiency of algorithms, big-O notation; object oriented programming concept difference in approach from procedural programming;
2. **Introduction to Linux and Java programming environment, Java compiler and virtual machine:** structure of a Java program, stand-alone programs and applets: concepts of portability;
3. **Basic Programming elements in Java:** data types, variables and array's operators, assignment and selection statements iterative structures, nested loops;
4. **Classes in Java:** general form of a class, creating objects, access control in classes. Constructors, methods, finalization, parameters, method overloading, recursive methods, returning objects, static members final, qualifier, nested and inner classes, string handling in Java, I/O mechanism, command line arguments.
5. **Inheritance:** basics super classes and subclasses, multilevel hierarchy, method overriding: run time polymorphism, abstract classes, final in inheritance, and the object class;
6. **Packages and Interfaces:** defining package, access protection, importing classes and packages, defining and implementing interfaces, nested interfaces, use of interfaces, variables in interfaces, the keyword extends;
7. **Exception handling :** fundamentals , types of exceptions catching exceptions, multiple catching nested try statements, uncaught exceptions, throw and throws, finally mechanism, built-in exceptions, creating exception subclasses, using exceptions;
8. **Applets :** applet fundamentals, native methods, static import, the applet class, applet display method, requesting repainting, a banner applet, passing parameters to applets, uses of applets.

References

1. Dietel and Associates, **Java How to Program**, 7th Edition, Prentice- Hall.
2. David F., **Java in a Nutshell**, 5th Edition, O'Reilly Media, Inc.,
3. David Gries, **The Science of Programming**
4. Dijkstra E. W., **A Short Introduction to the Art of Programming**
5. Dromey, **How to solve it by Computer**
6. Bruce E., **Thinking in Java**, Prentice Hall
7. Gosling J., Joy B., Steele G. and Bracha G., **The Java Language Specification**, 2nd Edition.